A regular four-point orbit at the crossing anchor

Human Compression project: bounded L044 proof component

Retain the accepted unique-crossing string C-group setting in $G \leq A_n$, $n \geq 12$, with all maximal omissions intransitive. The two G_i-orbits A, B have two blocks of sizes k, l. Assume $\alpha = 0$, $\beta = 1$, $\gamma = 0$, and that the crossing endpoint $a \in A$ lies in the regular four-point orbit D of $M \cong C_2^2$. All other M-orbits are pairs. Put $s = \rho_i$ and let $b \in B$ be the other crossing endpoint.

Lemma 1. *The stated configuration is impossible if $k \geq 6$ and $l \geq 4$.*

Proof. The two independent block-image generators must be the literal neighbors of i, since every other generator fixes both crossing endpoints. On B they cannot both exchange the blocks: they would then act identically on each of its paired rows. Their distant neighbors, each commuting with the opposite swap, and every other generator would fix the point opposite b. Its pair would be G_i-invariant. After reversal, therefore, write $\tau_L = \rho_{i-1}$, $\tau_R = \rho_{i+1}$, where τ_L fixes B pointwise and τ_R exchanges its blocks. Independence forces τ_L to exchange the blocks of A. The generator τ_R may either fix or exchange those blocks; both literal possibilities will be considered.

Both k, l are even. To see this, choose in the group M the element whose block image exchanges only A. On D it has two transpositions, and on the other paired rows of A it has $k-2$ transpositions; its sign is $(-1)^k$. The element whose block image exchanges only B has two transpositions on D and l on B, and sign $(-1)^l$. Both elements belong to A_n. This parity argument does not replace either distinguished generator in the marking.

Let $x = \rho_{i-2}$ and $y = \rho_{i+2}$ when present, and let C be all generators except these, τ_L, s, τ_R. They centralize M. Since they fix a, they fix the whole regular M-orbit D pointwise. Their action on the other paired rows is diagonal; they also fix the row containing b. Thus their faithful row group is contained in

$$S_{k-2} \times S_{l-1}.$$

Its independent length is at most $K-5$, where $K = k+l$. The exact deficit formula gives $r = K$, so there are at least $K-5$ generators in C, with equality only if both x, y exist. Consequently both exist and the equality clause gives the full row product above, with a maximal independent marking.

The prefix P of C and its suffix Q commute and have trivial intersection by IP. The two symmetric factors are centerless and directly indecomposable, so they distribute as whole factors between these intervals. The factor S_{l-1} must belong to Q: if it belonged to P, y would centralize its whole natural action, fixing the opposite point to b; all other generators except τ_R fix it as well, and τ_R only exchanges the anchor pair. This contradicts transitivity on B.

The other factor cannot also lie in Q. In that case x centralizes the entire row group, and so fixes B pointwise and all points of $A \setminus D$. On D, it preserves the two blocks, fixes a, and centralizes τ_R. If τ_R exchanges the blocks of A, this also fixes its opposite point and hence all four points of D. If τ_R fixes the blocks, its two transpositions on D make x either the identity or the single transposition on the block not containing a. The first alternative makes x trivial, the second odd. Thus $P = S_{k-2}$ and $Q = S_{l-1}$.

The generator y centralizes P and τ_L, preserves both blocks of A, and fixes a. It fixes every point outside D because the corresponding natural symmetric centralizers are trivial. It also fixes a^{τ_L}, and hence all four points of D. Therefore y is the identity on A.

If τ_R fixes the blocks of A, its support on A is exactly D. The generator x commutes with it and therefore preserves D. The same is already true of M, C, y, contradicting transitivity on A, since $k \geq 6$. It remains to consider the mixed basis, in which τ_R also exchanges the blocks of A.

If $l \geq 6$, the maximal independent string of the natural $Q = S_{l-1}$ is ordinary Coxeter; the outer degree-six exception cannot occur because $l-1$ is odd. Its near-end omission is S_{l-2} fixing the anchor and one additional row. Since y centralizes this omission, fixes b, and must connect the opposite anchor to another row, its action on B is one transposition. It is trivial on A and so odd, impossible. Thus only $l = 4$ remains.

Write p, q for the far and near generators of Q, so they occur in the order s, τ_R, y, q, p. In row labels on B they are

$$p = (1\ 2)(1'\ 2'), \qquad q = (2\ 3)(2'\ 3').$$

Since y is trivial on A, the accepted four-row centralizer calculation gives exactly $y = (1\ 2)(0'\ 3')$ or $(1'\ 2')(0'\ 3')$. Moreover s commutes with Q, y and therefore fixes $B \setminus \{b\}$ pointwise.

Consider the endpoint omission of the maximal prefix marking of $P = S_{k-2}$. The accepted classifications, including the elementary degree-four cases, make this either a natural S_{k-3} on all but one of its rows, or a transitive subgroup on all $k-2$ rows. The latter includes the twisted S_4 omission D_8 and the outer S_6 omission. Since x centralizes this omission, that latter alternative would make x preserve $A \setminus D$, leaving D invariant under G_i. It is impossible. In the ordinary omission case, its centralizer on either large block is the symmetric group on the two rows of D and its one omitted row. The generator x fixes a and must connect D to the other rows, so on that block it is the transposition between the other row of D and the omitted row. Its action on the opposite block is determined by its commutation with τ_R.

It follows that $\langle P, x\rangle$ is the natural diagonal S_{k-1} on two orbits matched by τ_R, fixing exactly a, a^{τ_R} in A. Since s centralizes this group and sends a to b, it fixes a^{τ_R}. On the two remaining $(k-1)$-point orbits it either fixes every point or exchanges corresponding points by τ_R. The first option would make s the odd transposition $(a\ b)$, so the second is forced.

This is precisely the literal action used in the accepted four-row endpoint identity: the swap is τ_R, its four-row orbit is B, its even endpoint generator is y, and s exchanges all row pairs of A except the anchor pair, replacing that exchange by $(a\ b)$. Therefore

$$p = s(\tau_R s y)^5 \tau_R y.$$

The accepted ten-point cycle proof and cancelling extra row pairs apply without modification. It makes the distinguished generator p redundant, the final contradiction. □

Lemma 2 (A small central factor)**.** *Let $m \geq 3$ be odd. An independent involutory string C-marking of $C_2 \times S_m$ with m generators contains the central involution as a distinguished generator. The remaining generators form the graph of a character $S_m \to C_2$, with a maximal symmetric string marking.*

Proof. For $m \geq 5$, choose a minimal subfamily generating the S_m quotient. The accepted normal-quotient independent-length bound shows that exactly $m-1$ generators are needed, since the kernel contributes at most one. Their quotient images are an extremal involutory string, hence an ordinary Coxeter path by the accepted L047 classification; the even-degree outer exception is absent. Its noncommuting neighbors force the chosen subfamily to occupy consecutive indices in the original string. The sole remaining generator centralizes all but possibly one endpoint of this path. The endpoint omission's centralizer in S_m is trivial, so that generator is the central C_2 involution. IP makes the other subgroup intersect it trivially, proving the character-graph statement. This is the accepted small-factor proof, with its degree-five endpoint supplied by the same accepted classification.

For $m = 3$, write the central involution as z. All other involutions are $z^\epsilon t$ with t a transposition. If no distinguished generator is z, the commuting string ends have the same transposition

image. Being distinct, they are t and zt. The middle generator has a different transposition image. One of its two products with the ends has order six, so that pair already generates $C_2 \times S_3$, a contradiction. Thus z occurs. The other two generators project to distinct transpositions; their product must have order three, since otherwise they already generate the whole group. They therefore form the asserted complement. □

Lemma 3 (The remaining anchored block sizes)**.** *The regular-M-orbit configuration in this note is impossible for all $k, l \geq 2$ with $n \geq 12$.*

Proof. The parity argument above still gives even k, l. The paired anchor orbit in B still forces y to exist and to move the point opposite b. If $l = 2$, its action on B is consequently a single transposition: it fixes the first two-point block and exchanges the second. On A, it preserves the blocks and commutes with τ_L, which exchanges them. Its action there is diagonal and even. Thus y is odd, impossible. Hence take $l \geq 4$.

Suppose $k = 2$, so $A = D$ has four points. The generator x, if present, is the identity on A. Indeed, if τ_R exchanges its blocks, commutation and fixation of a fixes one point in each two-point block and thus all four points. If τ_R fixes those blocks, x can at most exchange the second block; on B it is diagonal by commutation with τ_R, so evenness again forces its four-point action to be the identity. The whole left interval, consisting of x and all earlier generators, is therefore supported on B and fixes b. It commutes with the other generators of G_i: the only possible exception in the string is τ_L, whose support is A, so they commute in fact. Since G_i is transitive on B, this left interval is trivial. In detail its commuting complementary factor has the whole orbit of b, so any element of the left factor fixes every point in that orbit. Consequently x and the left interval are absent.

Counting now gives a maximal right interval $Q = S_{l-1}$ on the nonanchor rows of B. The generator y is trivial on A, since it centralizes τ_L, preserves the two blocks and fixes a. For $l \geq 6$ the ordinary endpoint omission again makes it odd. For $l = 4$, its two even possibilities are exactly the four-row endpoint alternatives, and s fixes all seven nonanchor points of B. On $A \cup \{b\}$ it is the crossing transposition $(a\ b)$ times one transposition on the three remaining points of A. Choose coordinates so that the nonidentity involution $\tau_R|_A$ is $(A_0\ A_0')(A_1\ A_1')$ and $a = A_0$. These are precisely the six cases already proved in the accepted two-row endpoint: its words $w = p$ or $(\tau_R s \tau_R w)^3 = p$ make the far generator of Q redundant. No change of distinguished generators is involved in this coordinate relabeling.

Finally suppose $k = 4$. The independent-length count used above again forces both x, y to exist and gives the full row group

$$\langle C \rangle = C_2 \times S_{l-1}.$$

Its central involution z exchanges the two unused rows of A diagonally; its support is exactly $A \setminus D$. Put $m = l - 1$, which is odd. The preceding lemma applies to the maximal rank-m marking of C.

The noncentral S_m component must lie in the right interval: otherwise y centralizes its entire natural action on each block of B, fixing the opposite anchor. If the central generator z also lies in the right interval, its final generator adjacent to y must be in the S_m component, for the same reason. Thus y commutes with z, while x centralizes all of C. Both preserve its support $A \setminus D$; the remaining generators preserve D as well, a contradiction. Therefore the left interval is the single distinguished generator z, and the right interval is a character graph over S_m.

The character graph is trivial. If it were nontrivial, commutation of x with an odd graph element would force $x|_A$ to centralize $z|_A$ (restrict the commutator to A). Hence x would preserve D. Also y commutes with z and so preserves D, contradicting transitivity on A again. We thus have a pure right-hand S_m factor.

The generator y centralizes z and τ_L, preserves the blocks and fixes a. It fixes D pointwise, and on its complement is either the identity or z. In particular its action on A is even. For $l \geq 6$ its action on B is the single odd endpoint transposition, again impossible. Take $l = 4$, and

write $y|_A = z^\eta$ with $\eta \in \{0, 1\}$. The element $y_0 = yz^\eta$ is supported on B, with one of the two even four-row endpoint actions. It is a word in available generators, not a replacement marking. The generator s commutes with z, y and the pure right interval, hence with y_0; it fixes $B \setminus \{b\}$ pointwise.

The pure-block possibility for $\tau_R|_A$ is impossible, since its support D would be preserved by x, which centralizes it. In the mixed possibility τ_R exchanges the blocks of A. The generator x fixes a and centralizes this swap. On the three other points of its first block it must exchange the remaining point of D with one of the unused rows, to connect D to its complement; this determines its action on the opposite block. Hence $\langle z, x\rangle$ is a diagonal S_3 on two orbits matched by τ_R, fixing a, a^{τ_R}. Centralization and evenness now force s to exchange those two three-point orbits by τ_R, fixing a^{τ_R} and exchanging a, b. The accepted four-row identity therefore gives

$$p = s(\tau_R s y_0)^5 \tau_R y_0,$$

where p is the far generator of the right interval. Its right side uses only distinguished generators other than p, making p redundant. This proves the final small case. □

Unfinished scope. This proof addresses the case where the regular M-orbit contains its crossing endpoint. Regular orbits away from that endpoint remain separate, as does the $\beta = 0$, $\gamma = 1$ branch and the other source obligations. The pure and mixed marked bases are distinguished throughout. This is not a whole L044 recovery or a new-question claim.

Attribution and paid dependencies. The source problem and orbit-cut framework are from Cameron, Fernandes, Leemans and Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Section 5.1. The present marked-basis and endpoint arguments are new proofs of bounded cases in that framework. They use the complete accepted block-deficit, small-endpoint and two-row notes, and the full retained L047 extremal/normal-quotient interfaces, including Whiston's independent-generation bounds. Their exact files and accepted reviews are bound in the companion dependency map; no sibling proof is an uncharged premise. No new finite census or formal theorem is used.